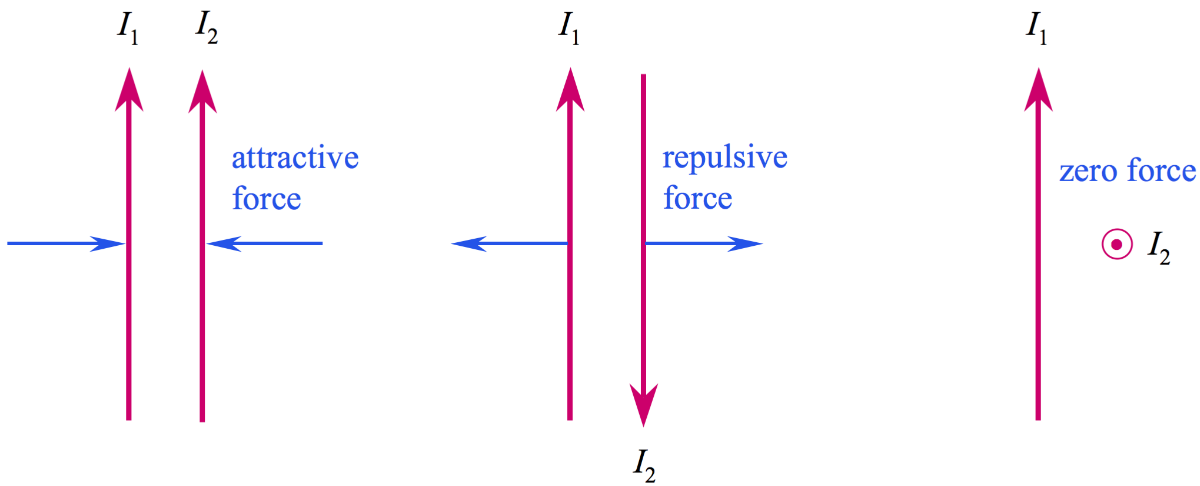



Engineering Electromagnetics
工程电磁场

Youwei Jia 贾有为

11: The Steady Magnetic Field

- Motivating the Magnetic Field Concept: Forces Between Currents



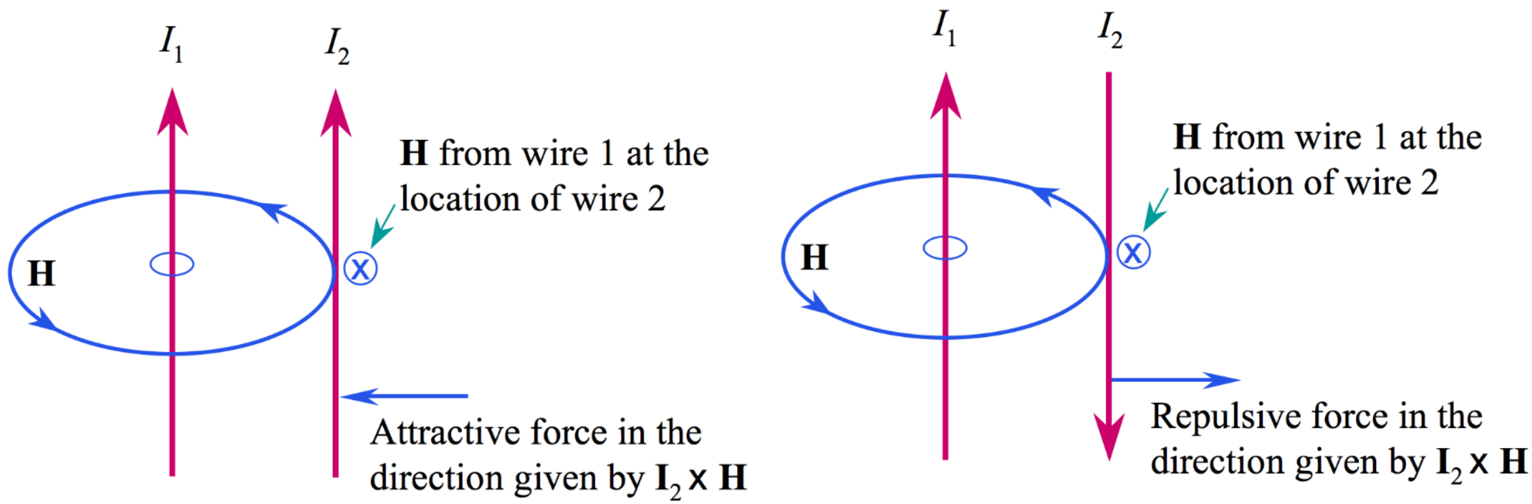
Magnetic forces arise whenever we have charges in motion 运动的电荷!

The source of the steady magnetic field may be a permanent magnet 永磁体, an electric field changing linearly with time随时间线性变化的电场, or a direct current直流电流. We will largely ignore the permanent magnet and save the time-varying electric field for a later discussion. Our present study will concern the magnetic field produced by a differential dc element in free space.

Q: What is the source of electrostatic field?

11: The Steady Magnetic Field

- Motivating the Magnetic Field Concept: Forces Between Currents



The magnetic field intensity, **H**, circulates around its source, I_1, in a direction most easily determined by the right-hand rule: Right thumb in the direction of the current, fingers curl in the direction of **H**.

11: The Steady Magnetic Field

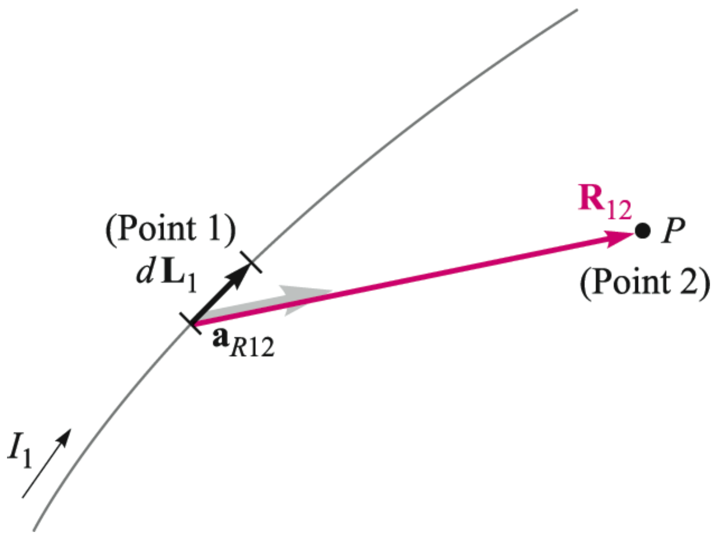
- ## Biot-Savart Law

The Biot-Savart Law specifies the magnetic field intensity, **H**, arising from a "point source" current element of differential length d**L**.


A. M. Ampère (1775-1836)


J. B. Biot (1774-1862)


F. Savart (1791-1841)



$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$

The units of **H** are [A/m]

Note in particular the inverse-square distance dependence, and the fact that the cross product will yield a field vector that points into the page. This is a formal statement of the right-hand rule.

Note the similarity to Coulomb's Law, in which a point charge of magnitude dQ_1 at Point 1 would generate electric field at Point 2 given by:

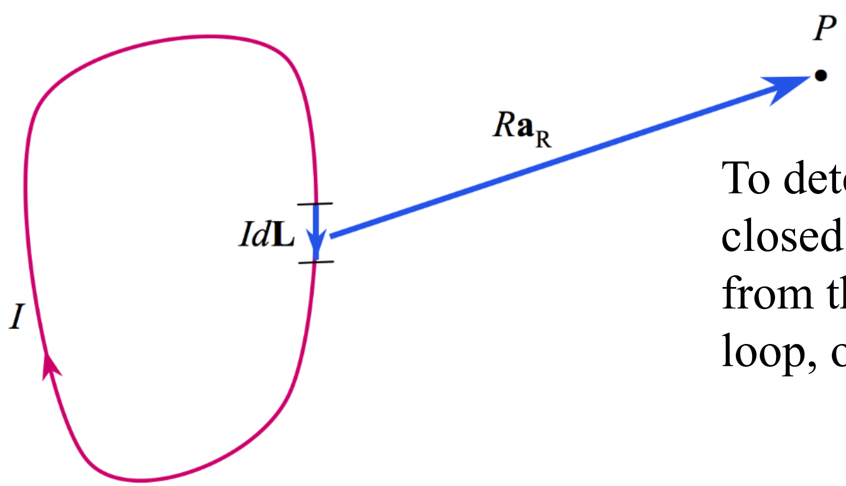
$$d\mathbf{E}_2 = \frac{dQ_1 \mathbf{a}_{R12}}{4\pi\epsilon_0 R_{12}^2}$$

11: The Steady Magnetic Field

- Magnetic Field Arising From a Circulating Current

At point P, the magnetic field associated with the differential current element $I d\mathbf{L}$ is

$$d\mathbf{H} = \frac{I\,d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$



To determine the total field arising from the closed circuit path, we sum the contributions from the current elements that make up the entire loop, or

$$\boxed{\mathbf{H} = \oint \frac{I\,d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}}$$

The contribution to the field at P from any portion of the current will be just the above integral evaluated over just that portion.

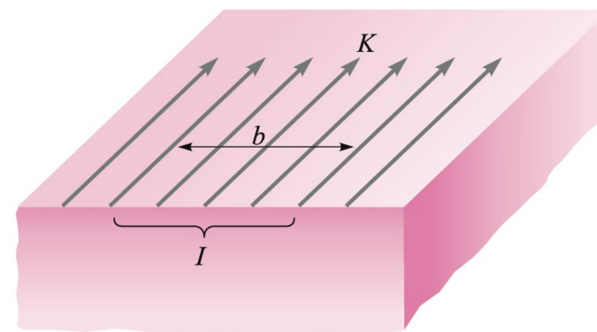
11: The Steady Magnetic Field

- Two- and Three-Dimensional Currents

$$d\mathbf{H} = \frac{I\,d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

On a surface that carries uniform surface current density $\mathbf{K}$ [A/m], the current within width b is

$$I = Kb$$



..and so the differential current quantity that appears in the Biot-Savart law becomes:

$$I\,d\mathbf{L} = \mathbf{K}\,dS$$

The magnetic field arising from a current sheet is thus found from the two-dimensional form of the Biot-Savart law:

$$\mathbf{H} = \int_S \frac{\mathbf{K} \times \mathbf{a}_R\,dS}{4\pi R^2}$$

In a similar way, a **volume current** will be made up of three-dimensional current elements, and so the Biot-Savart law for this case becomes:

$$\mathbf{H} = \int_{\mathrm{vol}} \frac{\mathbf{J} \times \mathbf{a}_R\,dv}{4\pi R^2}$$

• Example of the Biot-Savart Law

In this example, we evaluate the magnetic field intensity on the *y* axis (equivalently in the *xy* plane) arising from a filament current of infinite length on the *z* axis.

$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$

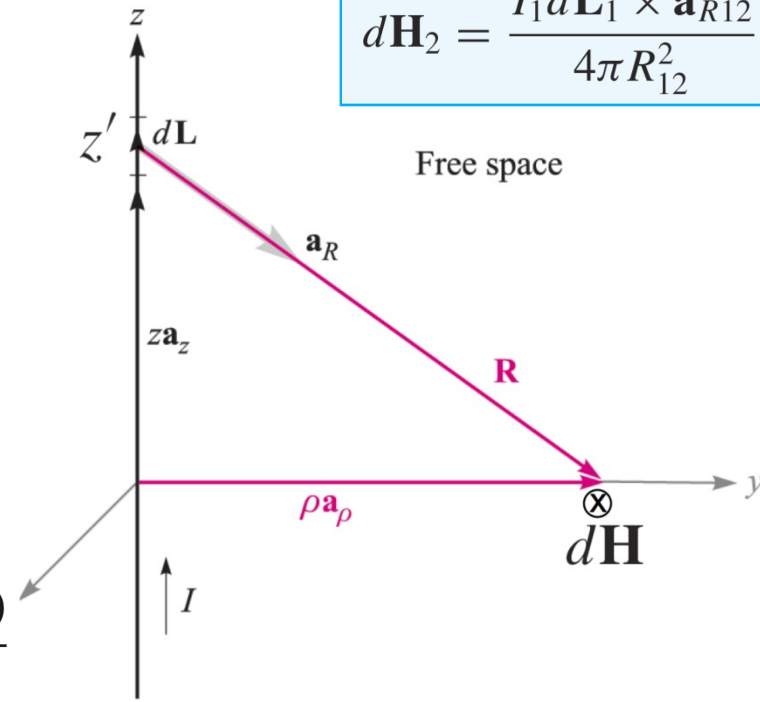
Using the drawing, we identify:

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = \rho\mathbf{a}_\rho - z'\mathbf{a}_z$$

and so.. $$\mathbf{a}_R = \frac{\rho\mathbf{a}_\rho - z'\mathbf{a}_z}{\sqrt{\rho^2 + z'^2}}$$

so that:

$$d\mathbf{H} = \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{I\, dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$



11: The Steady Magnetic Field

- Example of the Biot-Savart Law (Cont'd)

We now have:

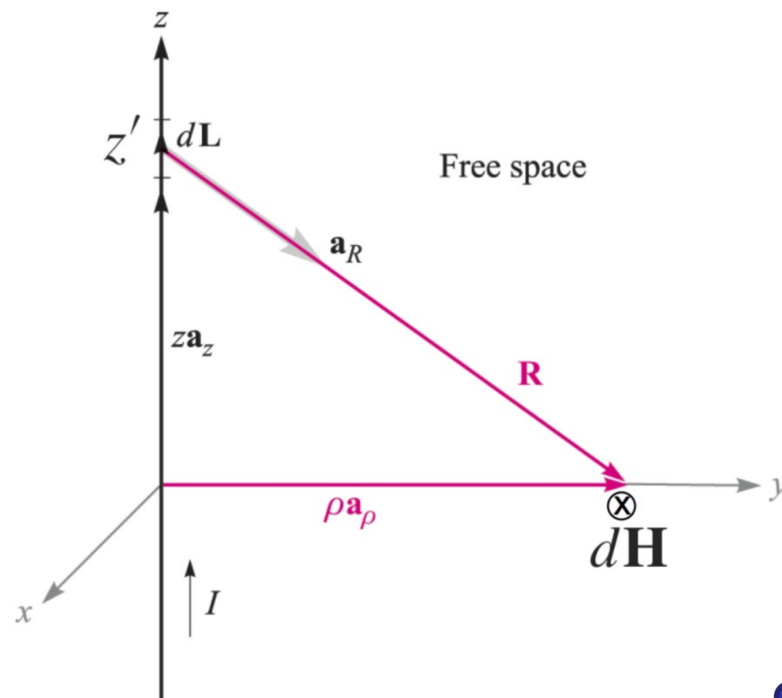
$$d\mathbf{H} = \frac{I\,d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

Integrate this over the entire wire:

$$\mathbf{H} = \int_{-\infty}^{\infty} \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

$$= \frac{I}{4\pi}\int_{-\infty}^{\infty} \frac{\rho\,dz'\mathbf{a}_\phi}{(\rho^2 + z'^2)^{3/2}}$$

..after carrying out the cross product

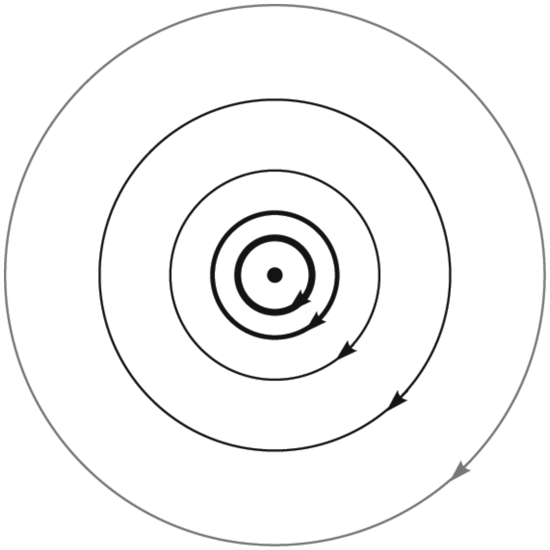


- Example of the Biot-Savart Law (Cont'd)

we have: $$\mathbf{H} = \frac{I}{4\pi}\int_{-\infty}^{\infty} \frac{\rho\, dz'\mathbf{a}_\phi}{(\rho^2 + z'^2)^{3/2}}$$

$$= \frac{I\rho\mathbf{a}_\phi}{4\pi} \frac{z'}{\rho^2\sqrt{\rho^2 + z'^2}}\Bigg|_{-\infty}^{\infty}$$

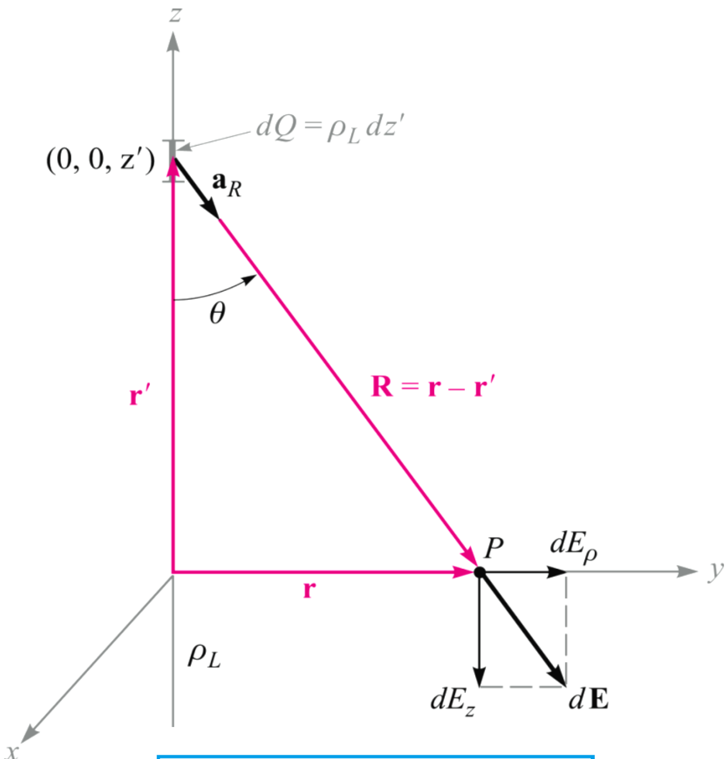
finally: $$\boxed{\mathbf{H} = \frac{I}{2\pi\rho}\mathbf{a}_\phi}$$

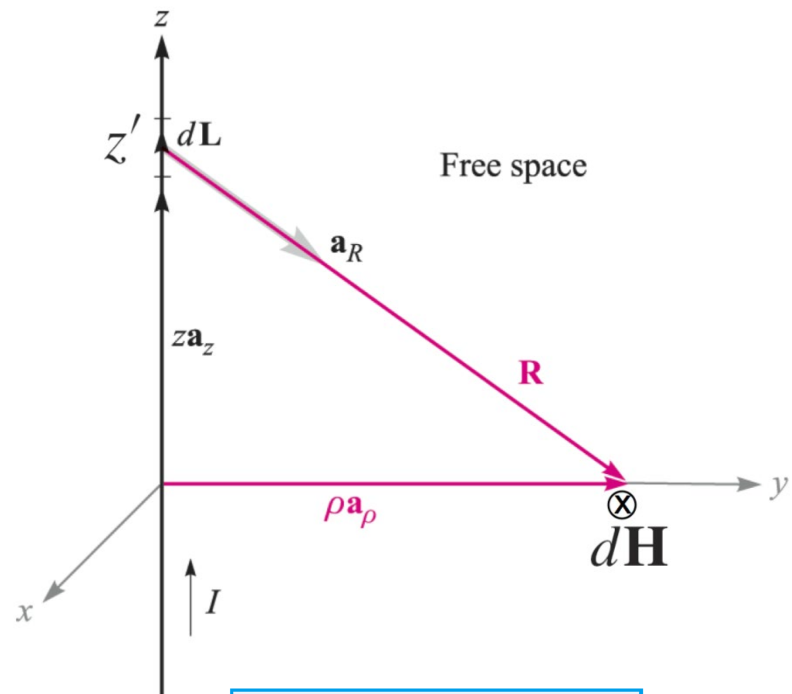


Current is into the page.
Magnetic field streamlines
are concentric circles, whose magnitudes
decrease as the inverse distance from the *z* axis

11: The Steady Magnetic Field

- Comparison: Electric field of a line charge Vs Magnetic field of a line current





$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$

$$\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi$$

11: The Steady Magnetic Field

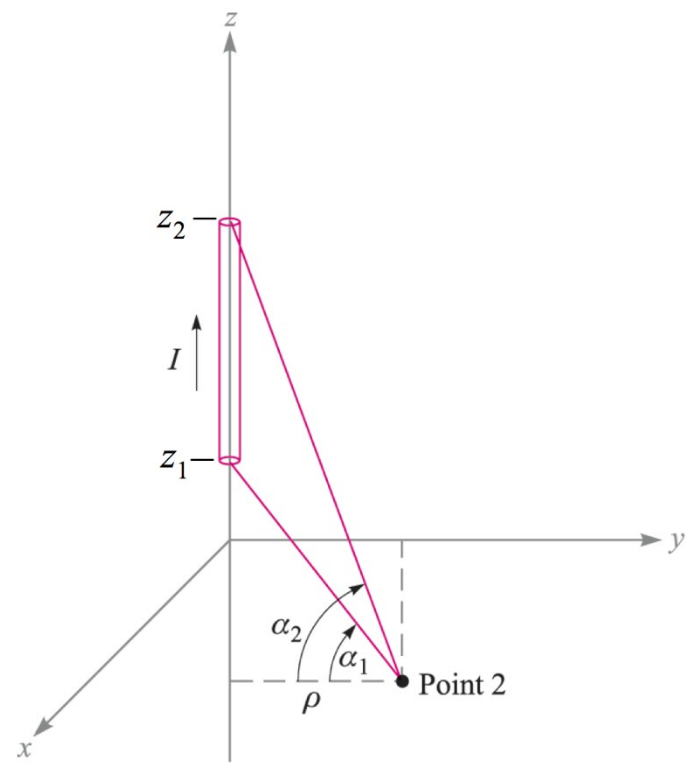
- Magnetic Field Arising from a Finite Current Segment

In this case, the field is to be found in the *xy* plane at Point 2. The Biot-Savart integral is taken over the wire length:

$$\mathbf{H} = \int_{z_1}^{z_2} \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

$$= \int_{\rho \tan \alpha_1}^{\rho \tan \alpha_2} \frac{I dz \mathbf{a}_z \times (\rho \mathbf{a}_\rho - z \mathbf{a}_z)}{4\pi(\rho^2 + z^2)^{3/2}}$$

$$= \frac{I\rho \mathbf{a}_\phi}{4\pi} \frac{z}{\rho^2 \sqrt{\rho^2 + z^{\ 2}}} \Bigg|_{\rho \tan \alpha_1}^{\rho \tan \alpha_2}$$



..after a few additional steps we find:

$$\boxed{\mathbf{H} = \frac{I}{4\pi\rho}(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\phi}$$

11: The Steady Magnetic Field

- Another Example: Magnetic Field from a Current Loop

Consider a circular current loop of radius a in the x-y plane, which carries steady current I. We wish to find the magnetic field strength anywhere on the z axis.

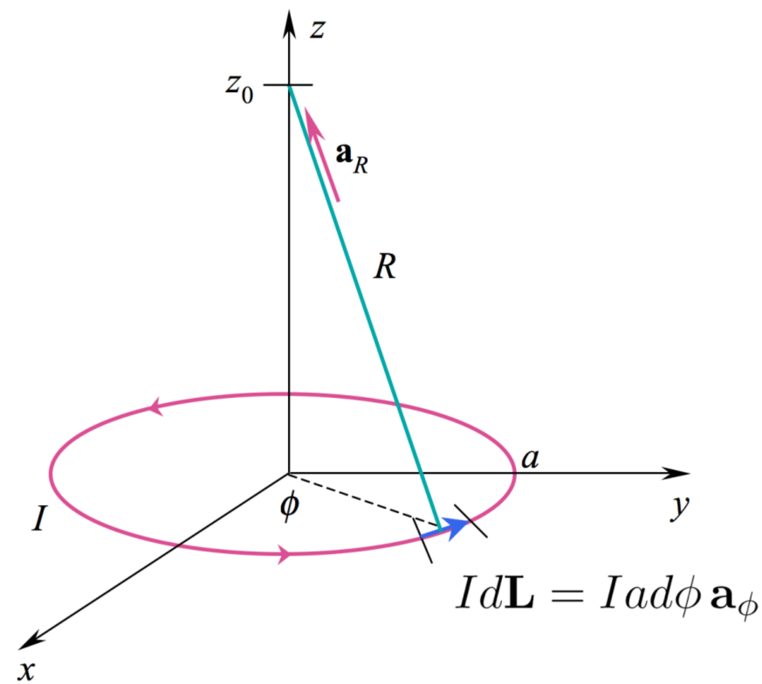
We will use the Biot-Savart Law:

$$\mathbf{H} = \int \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

where: $I d\mathbf{L} = I a d\phi \, \mathbf{a}_\phi$

$$R = \sqrt{a^2 + z_0^2}$$

$$\mathbf{a}_R = \frac{z_0 \, \mathbf{a}_z - a \, \mathbf{a}_\rho}{\sqrt{a^2 + z_0^2}}$$



11: The Steady Magnetic Field

- Another Example: Magnetic Field from a Current Loop

Substituting the previous expressions, the Biot-Savart Law becomes:

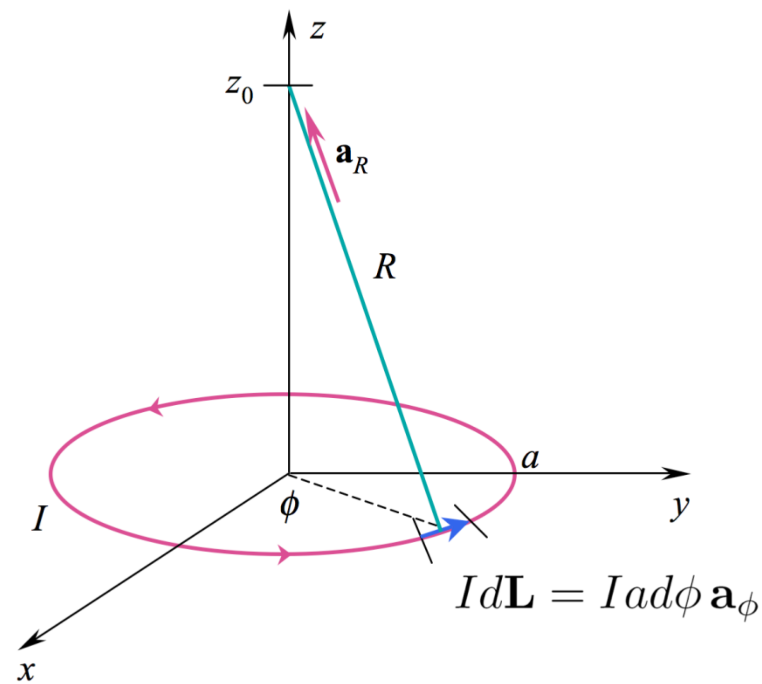
$$\mathbf{H} = \int_0^{2\pi} \frac{Iad\phi\, \mathbf{a}_\phi \times (z_0\, \mathbf{a}_z - a\, \mathbf{a}_\rho)}{4\pi(a^2 + z_0^2)^{3/2}}$$

carry out the cross products to find:

$$\mathbf{H} = \int_0^{2\pi} \frac{Iad\phi\, (z_0\, \mathbf{a}_\rho + a\, \mathbf{a}_z)}{4\pi(a^2 + z_0^2)^{3/2}}$$

but we must include the angle dependence in the radial unit vector:

$$\mathbf{a}_\rho = \cos\phi\, \mathbf{a}_x + \sin\phi\, \mathbf{a}_y$$



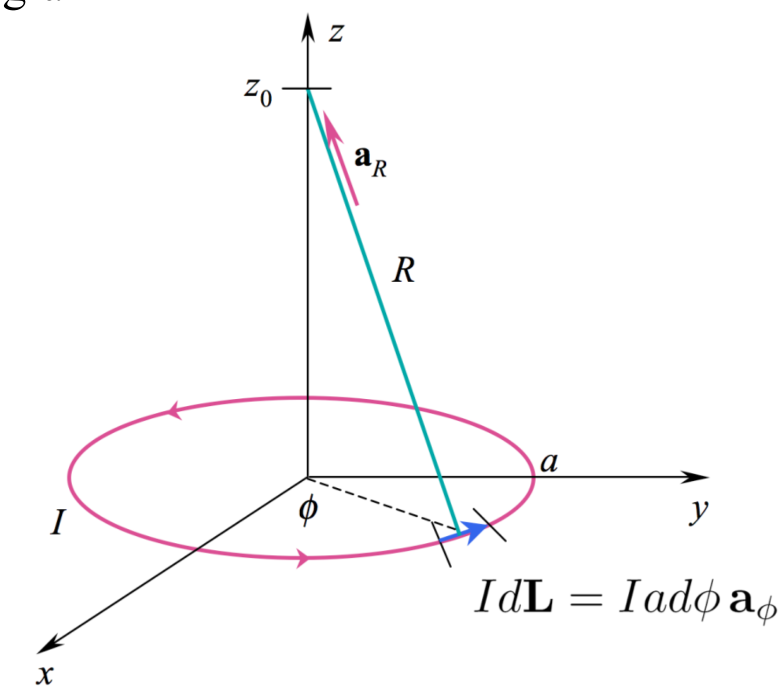
with this substitution, the radial component will integrate to zero, meaning that all radial components will cancel on the *z* axis.

11: The Steady Magnetic Field

- Another Example: Magnetic Field from a Current Loop

Now, only the z component remains, and the integral evaluates easily:

$$\boxed{\mathbf{H} = \frac{I(\pi a^2)\mathbf{a}_z}{2\pi(a^2 + z_0^2)^{3/2}}}$$



Note the form of the numerator: the product of the current and the loop area. We define this as the magnetic moment 磁矩:

$$\mathbf{m} = I(\pi a^2)\,\mathbf{a}_z$$

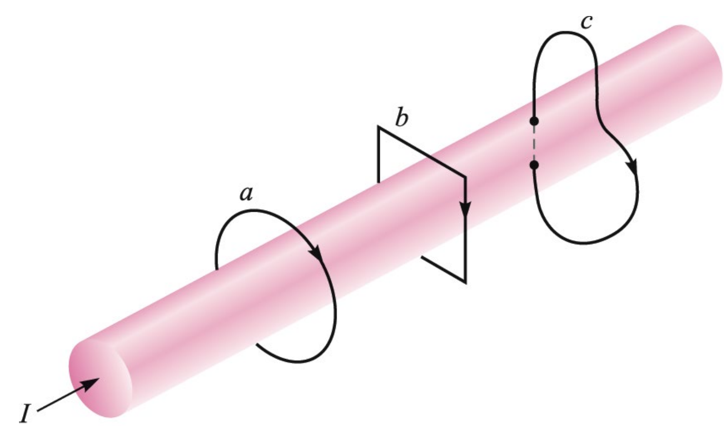
Note: Please compare the magnetic moment with the electric dipole moment which both are vectors.

11: The Steady Magnetic Field

- Ampere's Circuital Law 安培环路定律

Ampere's Circuital Law states that the line integral of **H** about any closed path is exactly equal to the direct current enclosed by that path.

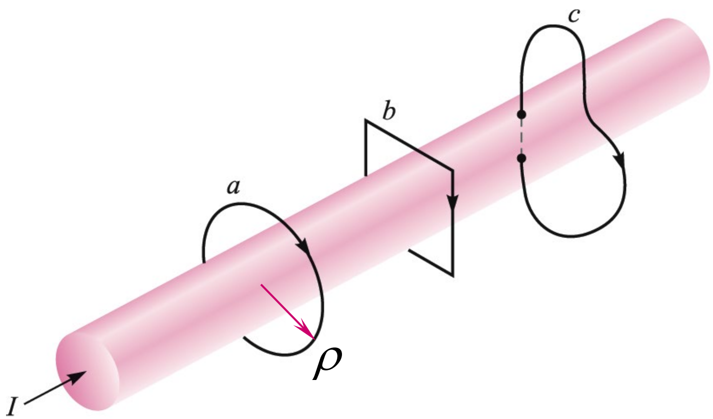
$$\boxed{\oint \mathbf{H} \cdot d\mathbf{L} = I}$$



In the figure at right, the integral of **H** about closed paths *a* and *b* gives the total current *I*, while the integral over path *c* gives only that portion of the current that lies within *c*.

11: The Steady Magnetic Field

- Ampere's Law Applied to a Long Wire



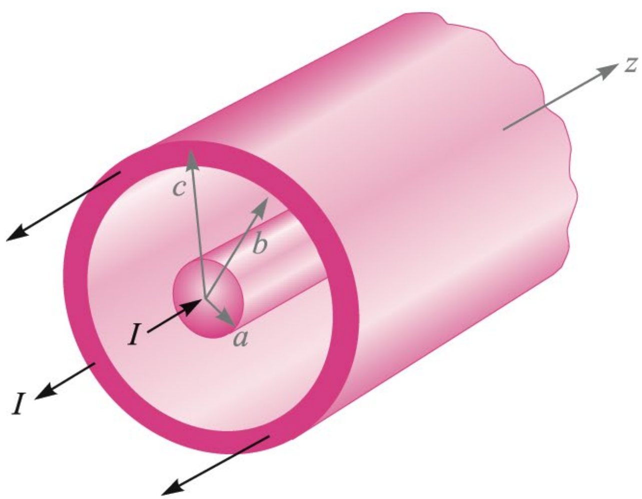
Symmetry suggests that **H** will be circular, constant-valued at constant radius, and centered on the current (z) axis.

Choosing path a, and integrating **H** around the circle of radius ρ gives the enclosed current, I:

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{2\pi} H_\phi \rho d\phi = H_\phi \rho \int_0^{2\pi} d\phi = H_\phi 2\pi \rho = I$$

so that: $\boxed{H_\phi = \dfrac{I}{2\pi\rho}}$ as before.

- Ampere's Law Applied to Coaxial Transmission Line



In the coax line, we have two concentric solid conductors that carry equal and opposite currents, *I*.

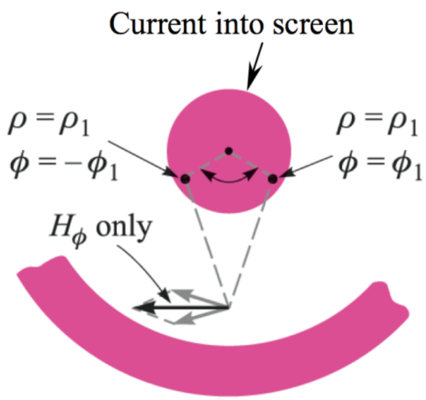
The line is assumed to be infinitely long, and the circular symmetry suggests that **H** will be entirely ϕ - directed, and will vary only with radius ρ.

Our objective is to find the magnetic field for all values of ρ .

11: The Steady Magnetic Field

- Ampere's Law Applied to Coaxial Transmission Line

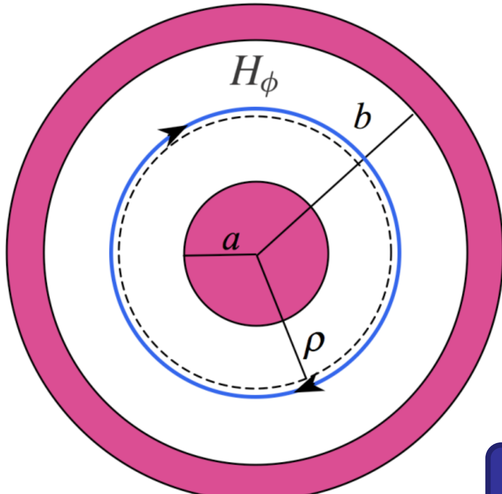
Field Between Conductors



The inner conductor can be thought of as made up of a bundle of filament currents, each of which produces the field of a long wire.

Symmetry suggests that **H** will be circular, constant-valued at constant radius, and centered on the current (z) axis.

Choosing the closed circle in blue with radius as ρ, and integrating **H** around this circle gives the enclosed current, I:



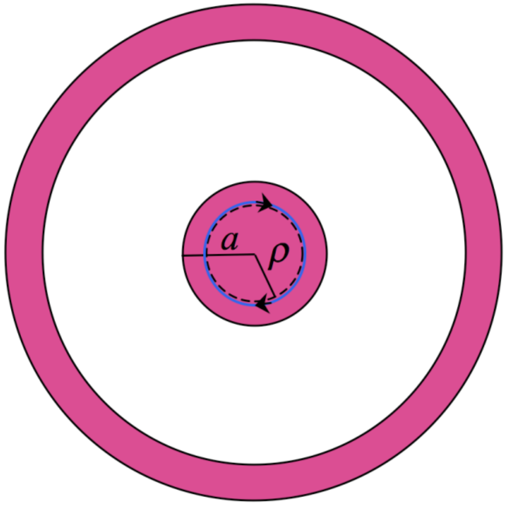
$$\boxed{H_\phi = \frac{I}{2\pi\rho} \qquad a < \rho < b}$$

- Ampere's Law Applied to Coaxial Transmission Line

Field Within the Inner Conductor

Inside the inner conductor, and at radius ρ, we again have:

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{2\pi} H_\phi \rho d\phi = H_\phi 2\pi \rho$$



But now, the current enclosed is

$$I_{\text{encl}} = I \frac{\rho^2}{a^2}$$

so that $2\pi\rho H_\phi = I \dfrac{\rho^2}{a^2}$ or finally: $\boxed{H_\phi = \dfrac{I\rho}{2\pi a^2} \quad (\rho < a)}$

11: The Steady Magnetic Field

- Ampere's Law Applied to Coaxial Transmission Line

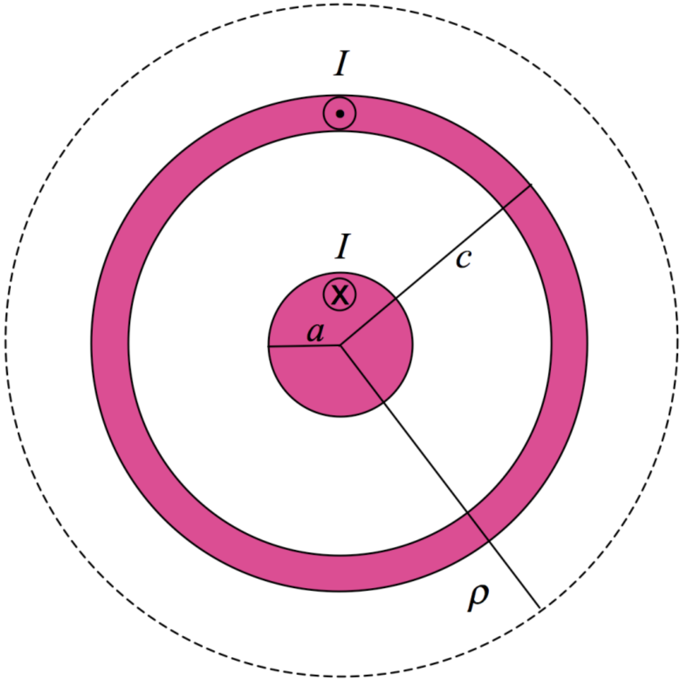
Field Outside Both Conductors

Outside the transmission line, where $\rho > c$, no current is enclosed by the integration path, and so

$$\oint \mathbf{H} \cdot d\mathbf{L} = 0$$

As the current is uniformly distributed, and since we have circular symmetry, the field would have to be constant over the circular integration path, and so it must be true that:

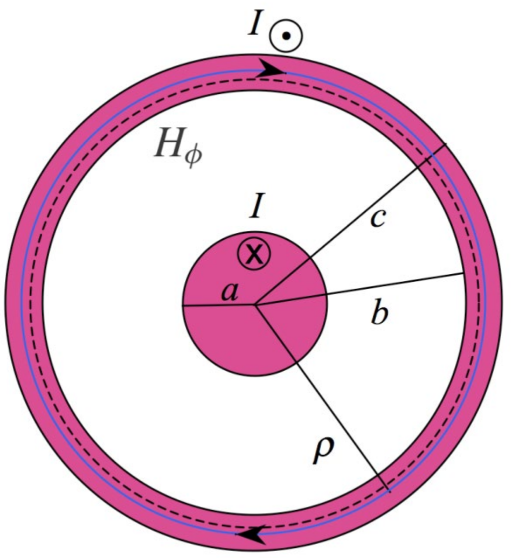
$$\boxed{H_\phi = 0 \quad (\rho > c)}$$



- Ampere's Law Applied to Coaxial Transmission Line

Field Inside the Outer Conductor

Inside the outer conductor, the enclosed current consists of that within the inner conductor plus that portion of the outer conductor current existing at radii less than ρ



Ampere's Circuital Law becomes

$$2\pi\rho H_\phi = I - I\left(\frac{\rho^2 - b^2}{c^2 - b^2}\right)$$

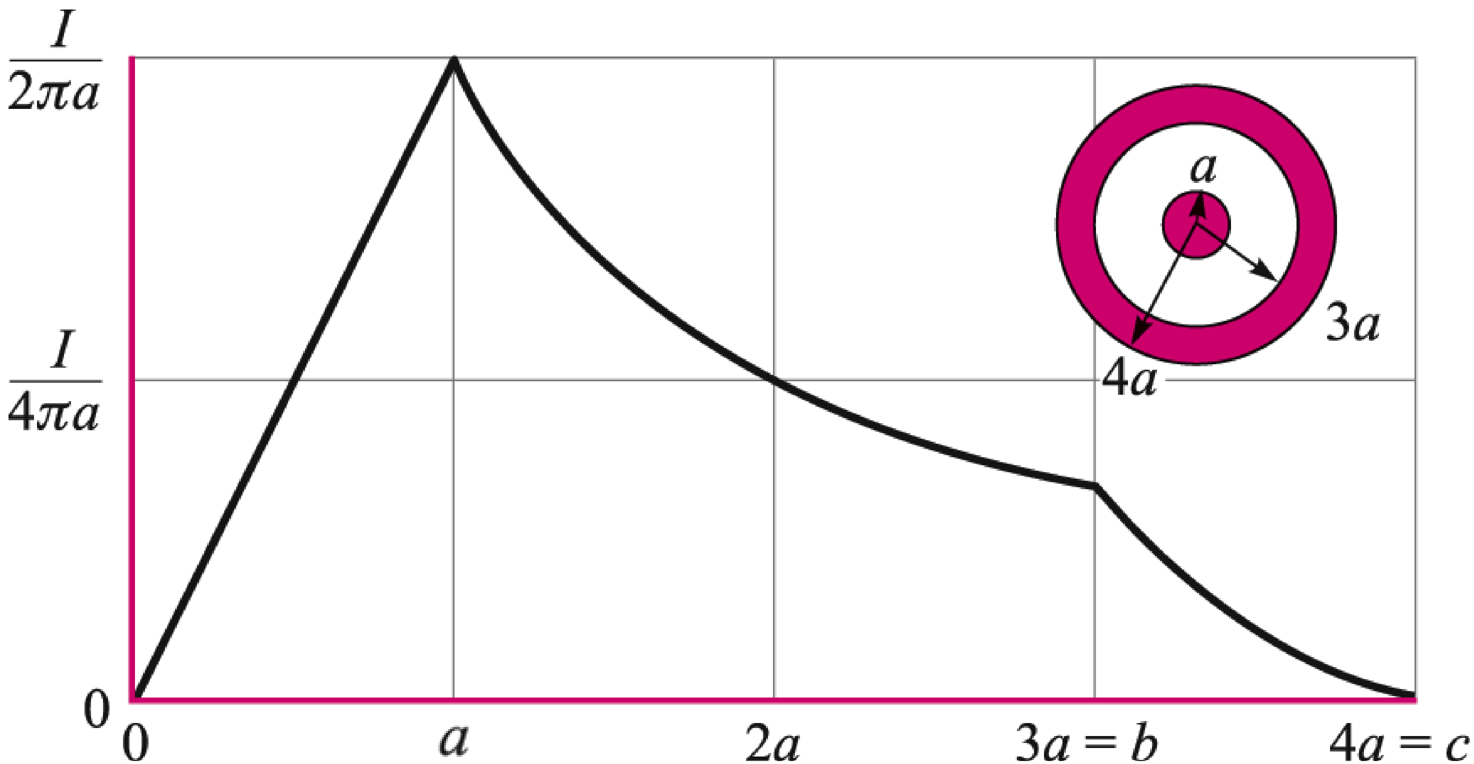
..and so finally:

$$\boxed{H_\phi = \frac{I}{2\pi\rho}\frac{c^2-\rho^2}{c^2-b^2} \quad (b < \rho < c)}$$

11: The Steady Magnetic Field

- Ampere's Law Applied to Coaxial Transmission Line

Magnetic Field Strength as a Function of Radius
in the Coaxial Transmission Line



11: The Steady Magnetic Field

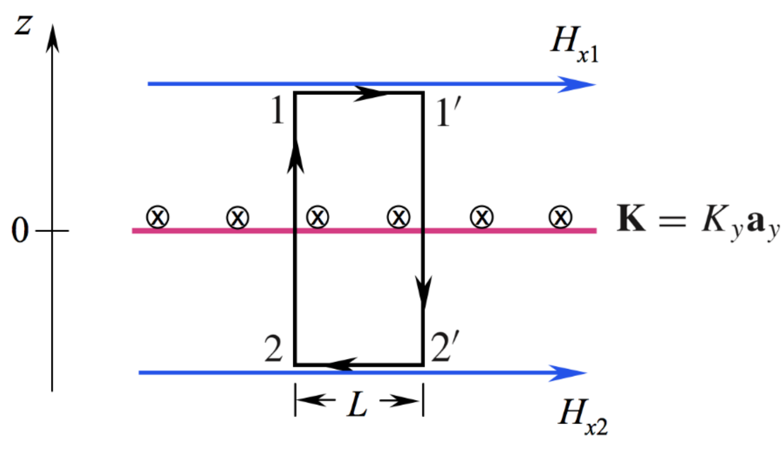
- Magnetic Field Arising from a Current Sheet

For a uniform plane current in the *y* direction, we expect an *x*-directed **H** field from symmetry.

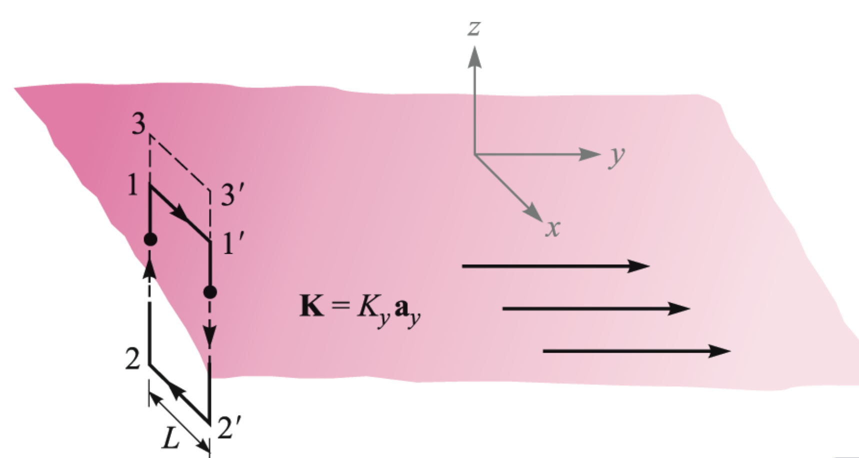
Applying Ampere's circuital law to the path $1\text{-}1'\text{-}2'\text{-}2\text{-}1$ we find:

$$H_{x1}L + H_{x2}(-L) = K_y L \quad \text{or} \quad H_{x1} - H_{x2} = K_y$$

In other words, the magnetic field is discontinuous across the current sheet by the magnitude of the surface current density.



edge view



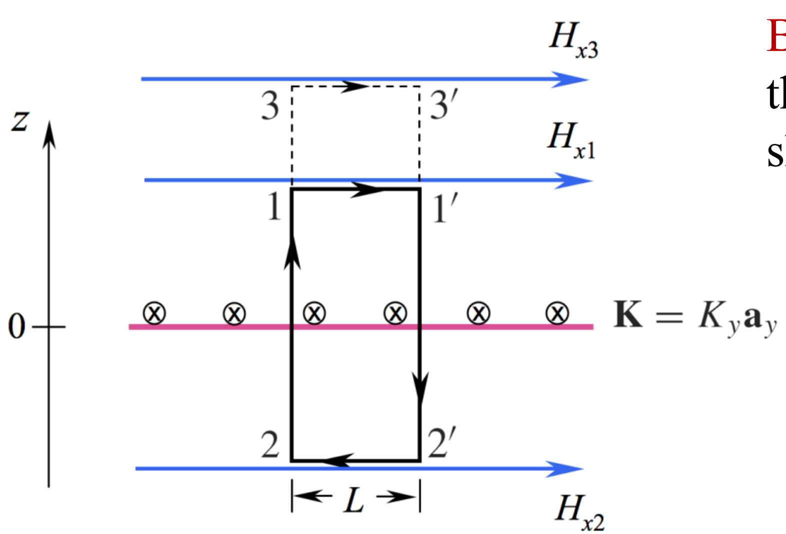
11: The Steady Magnetic Field

- Magnetic Field Arising from a Current Sheet

If instead, the upper path is elevated to the line between 3 and 3′ , the same current is enclosed and we would have

$$H_{x3} - H_{x2} = K_y \quad \text{from which we conclude that} \quad H_{x3} = H_{x1}$$

so the field is constant in each region (above and below the current plane)



edge view

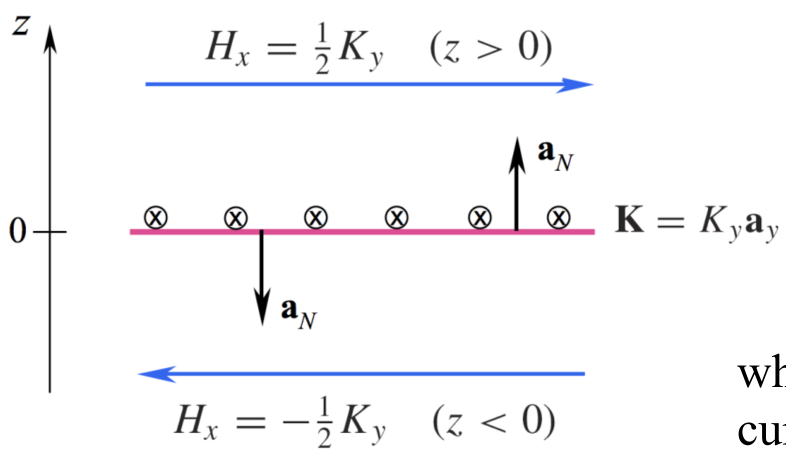
By symmetry, the field above the sheet must be the same in magnitude as the field below the sheet. Therefore, we may state that

$$\boxed{\begin{aligned} H_x &= \tfrac{1}{2} K_y \quad (z > 0) \\ \text{and} \quad H_x &= -\tfrac{1}{2} K_y \quad (z < 0) \end{aligned}}$$

- ## Magnetic Field Arising from a Current Sheet

The actual field configuration is shown below, in which magnetic field above the current sheet is equal in magnitude, but in the direction opposite to the field below the sheet.

The field in either region is found by the cross product:



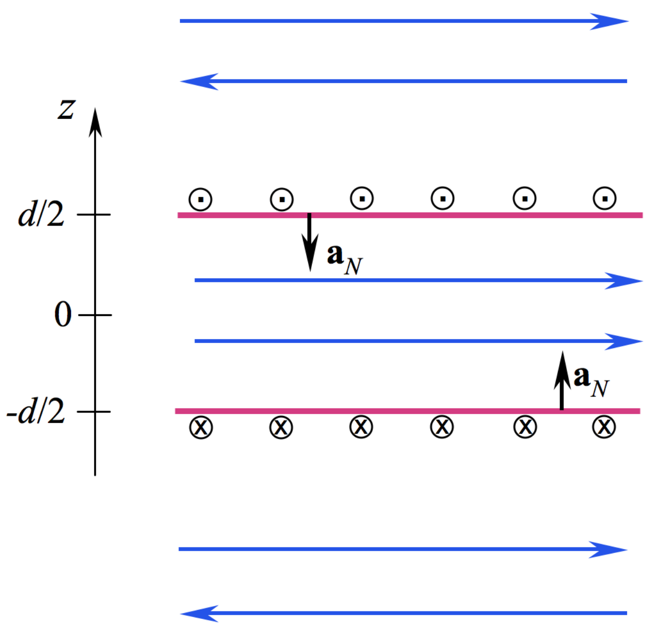
$$\boxed{\mathbf{H} = \frac{1}{2}\mathbf{K} \times \mathbf{a}_N}$$

where $\mathbf{a}_N$ is the unit vector that is normal to the current sheet, and that points into the region in which the magnetic field is to be evaluated.

11: The Steady Magnetic Field

• Magnetic Field Arising from Two Current Sheets

Here are two parallel currents, equal and opposite, as you would find in a parallel-plate transmission line. If the sheets are much wider than their spacing, then the magnetic field will be contained in the region between plates, and will be nearly zero outside.



$H_{x2}\,(z > d/2\,)$
$H_{x1}\,(z > d/2\,)$

These fields cancel for current sheets of infinite width.

$\mathbf{K}_1 = -K_y\,\mathbf{a}_y$

$H_{x1}\,(-d/2 < z < d/2\,)$
$H_{x2}\,(-d/2 < z < d/2\,)$

$\mathbf{K}_2 = -K_y\,\mathbf{a}_y$

These fields are equal and add to give

$$\boxed{\mathbf{H} = \mathbf{K} \times \mathbf{a}_N \quad (-d/2 < z < d/2\,)}$$

where **K** is either $\mathbf{K}_1$ or $\mathbf{K}_2$

$H_{x1}\,(z < -d/2\,)$
$H_{x2}\,(z < -d/2\,)$

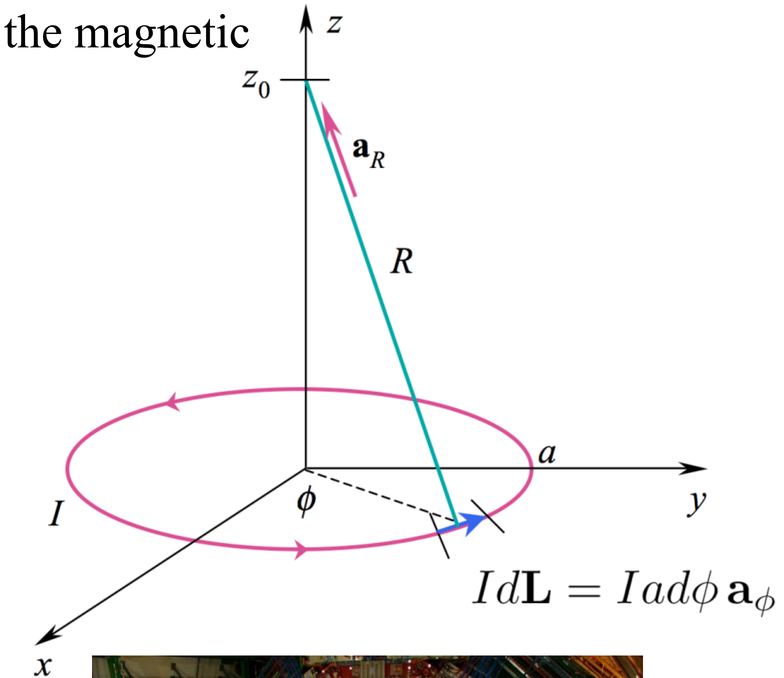
These fields cancel for current sheets of infinite width.

11: The Steady Magnetic Field

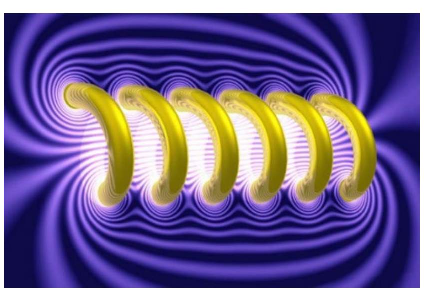
- Magnetic Field Arising from Current Loop Field

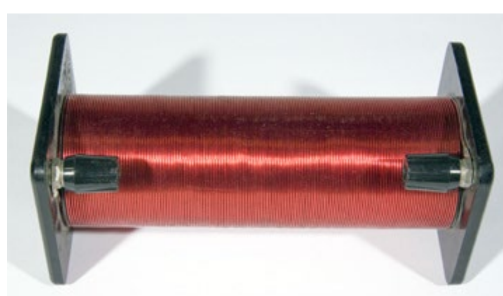
Using the Biot-Savart Law, we previously found the magnetic field on the *z* axis from a circular current loop:

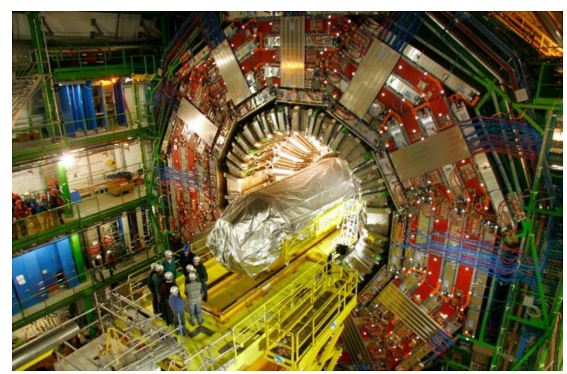
$$\mathbf{H} = \frac{I(\pi a^2)\mathbf{a}_z}{2\pi(a^2 + z_0^2)^{3/2}}$$



We will now use this result as a building block to construct the magnetic field on the axis of a solenoid 螺线管 -- formed by a stack of identical current loops, centered on the *z* axis.

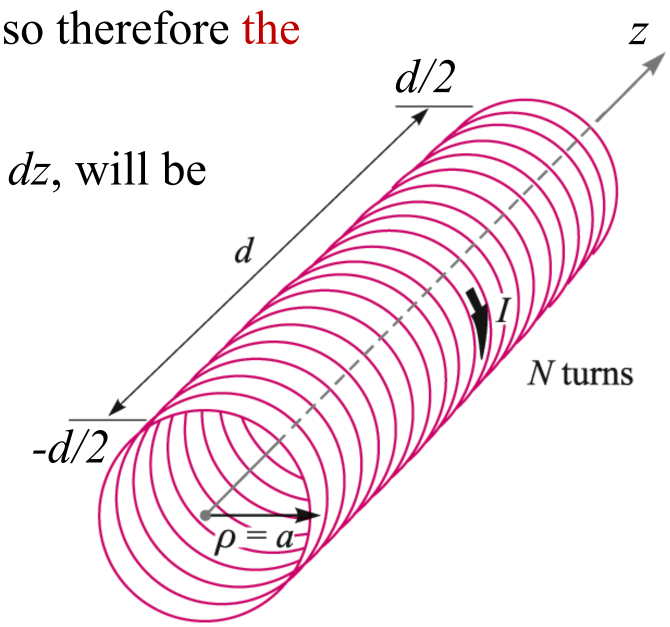






- On-Axis Field Within a Solenoid

We consider the single current loop field as a differential contribution to the total field from a stack of N closely-spaced loops, each of which carries current I. The length of the stack (solenoid) is d, so therefore the density of turns 匝数 will be N/d.



Now the current in the turns within a differential length, dz, will be

$$\boxed{dI = \frac{N}{d} I dz}$$

We consider this as our differential "loop current"

so that the previous result for **H** from a single loop:

$$\mathbf{H} = \frac{I(\pi a^2)\mathbf{a}_z}{2\pi(a^2 + z_0^2)^{3/2}}$$

now becomes:

$$d\mathbf{H} = \frac{(N/d)I dz(\pi a^2)\mathbf{a}_z}{2\pi(a^2 + z^2)^{3/2}}$$

in which z is measured from the center of the coil, where we wish to evaluate the field.

11: The Steady Magnetic Field

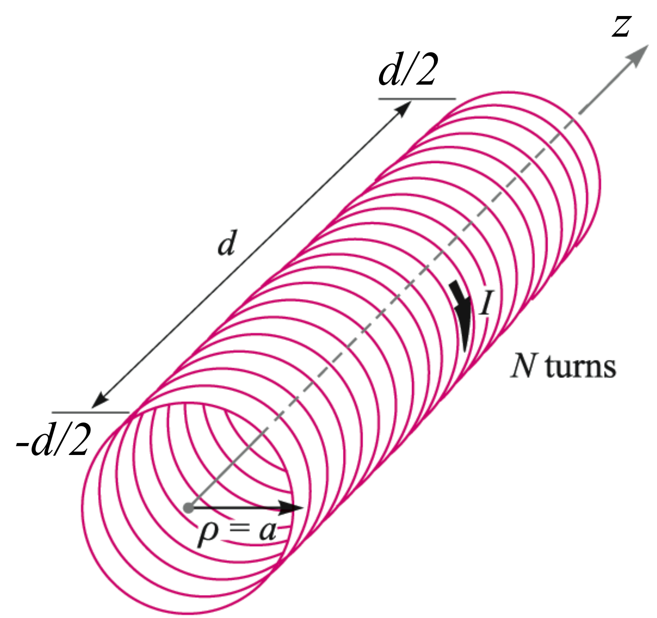
- ## On-Axis Field Within a Solenoid

The total field on the z axis at $z = 0$ will be the sum of the field contributions from all turns in the coil -- or the integral of $d\mathbf{H}$ over the length of the solenoid.

$$\mathbf{H} = \int d\mathbf{H} = \int_{-d/2}^{d/2} \frac{(N/d)I dz(\pi a^2)\mathbf{a}_z}{2\pi(a^2+z^2)^{3/2}}$$

$$= \frac{NIa^2}{2d}\mathbf{a}_z \int_{-d/2}^{d/2} \frac{dz}{(a^2+z^2)^{3/2}}$$

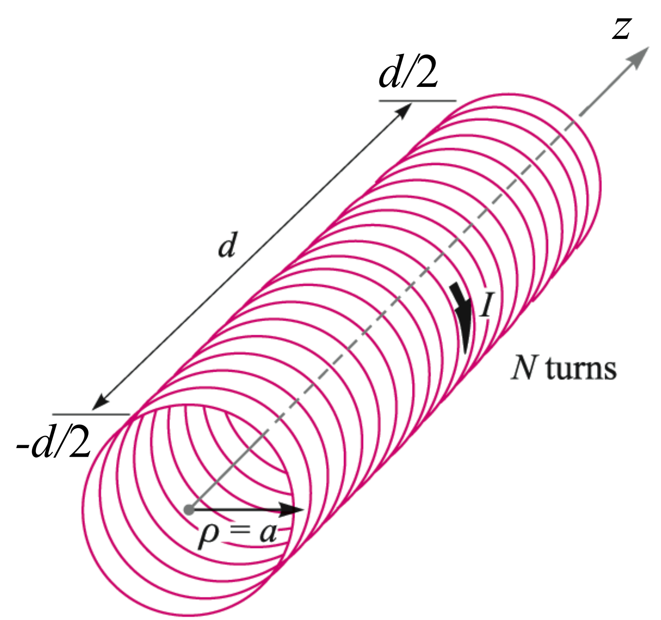
$$= \frac{NIa^2}{2d}\mathbf{a}_z \frac{d}{a^2\sqrt{a^2+(d/2)^2}} = \frac{NI\,\mathbf{a}_z}{2\sqrt{a^2+(d/2)^2}}$$



- On-Axis Field Within a Solenoid

We now have the on-axis field at the solenoid midpoint ($z = 0$):

$$\mathbf{H} = \frac{NI\,\mathbf{a}_z}{2\sqrt{a^2 + (d/2)^2}}$$



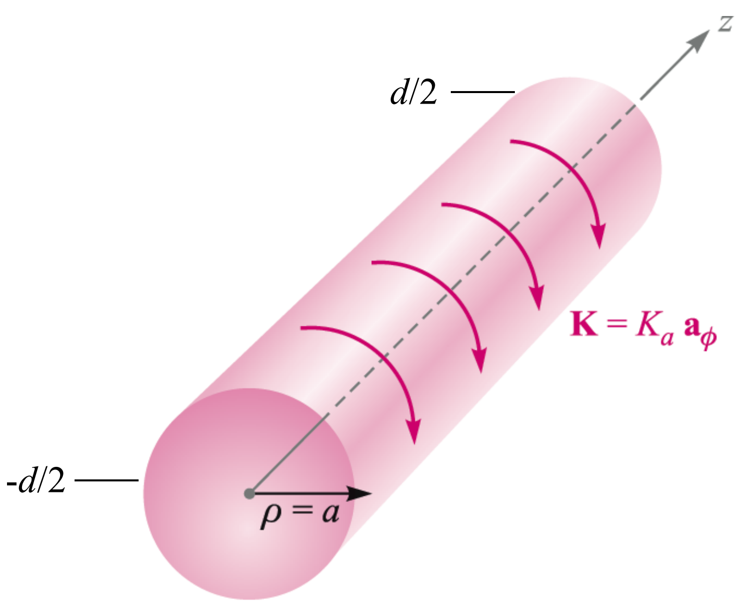
Note that for long solenoids, for which $d >> a$, the result simplifies to:

$$\mathbf{H} \doteq \frac{NI}{d}\,\mathbf{a}_z$$

This result is valid at all on-axis positions deep within long coils -- at distances from each end of several radii.

11: The Steady Magnetic Field

- On-Axis Field Within a Solenoid: Continuous Surface Current



$$\mathbf{K} = K_a \, \mathbf{a}_\phi = \frac{NI}{d} \, \mathbf{a}_\phi \quad \text{A/m}$$

Therefore:

$$\mathbf{H}(\rho = z = 0) = \frac{K_a d \, \mathbf{a}_z}{2\sqrt{a^2 + (d/2)^2}}$$

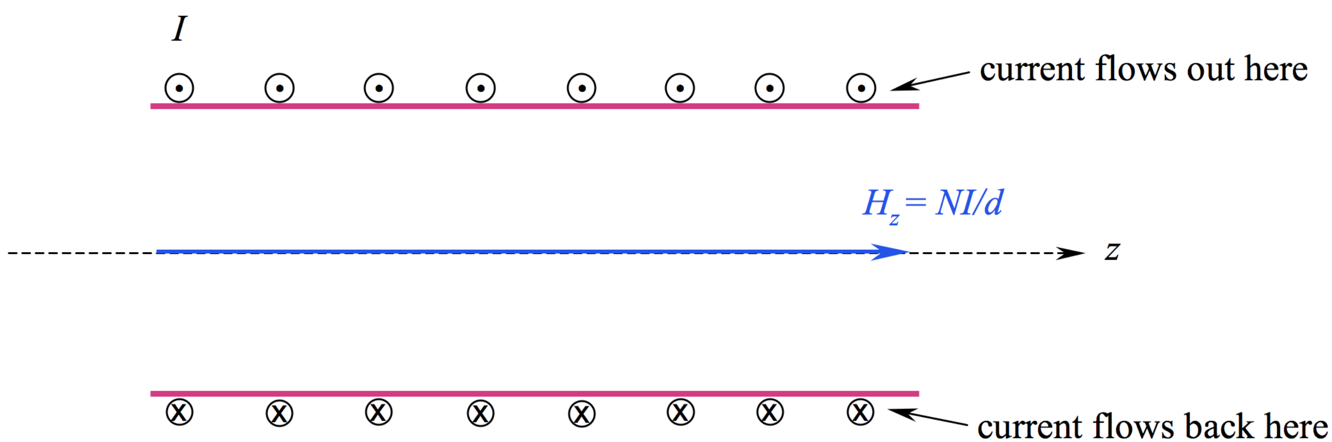
$$\doteq K_a \, \mathbf{a}_z \; (d >> a) \quad \text{A/m}$$

11: The Steady Magnetic Field

- ## Solenoid Field -- Off-Axis

To find the field within a solenoid, but off the *z* axis, we apply Ampere's Circuital Law in the following way:

The illustration below shows the solenoid cross-section, from a lengthwise cut through the z axis. Current in the windings flows in and out of the screen in the circular current path. Each turn carries current *I*. The magnetic field along the *z* axis is NI/d as we found earlier.

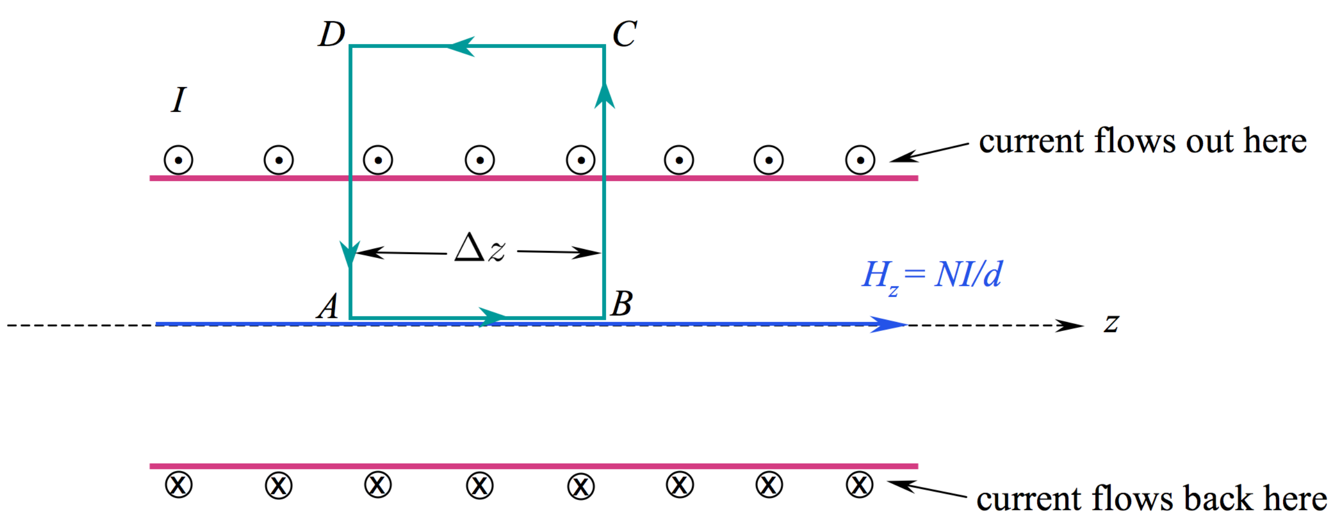


11: The Steady Magnetic Field

- Solenoid Field -- Off-Axis

Applying Ampere's Law to the rectangular path shown below leads to the following:

$$\oint \mathbf{H}\cdot d\mathbf{L} \;=\; \int_A^B H_z\,dz + \int_B^C H_\rho\,d\rho + \int_C^D H_{z,out}\,dz + \int_D^A H_\rho\,d\rho \;=\; I_{encl} \;=\; \frac{NI}{d}\Delta z$$

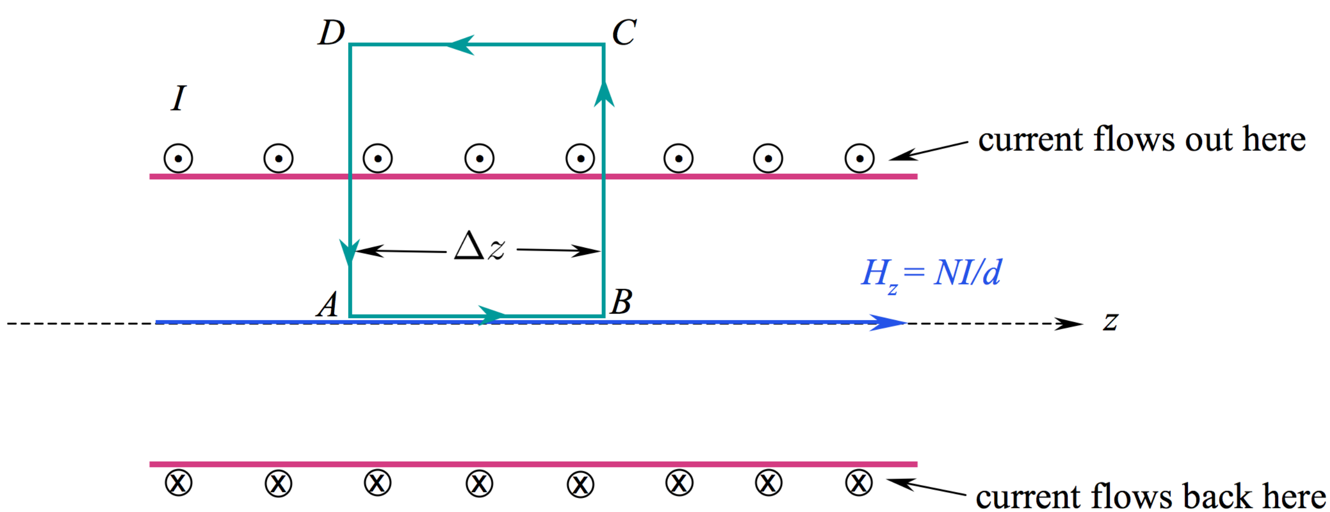


11: The Steady Magnetic Field

- ## Solenoid Field -- Off-Axis

The radial integrals will now cancel, because they are oppositely-directed, and because in the long coil, H_ρ is not expected to differ between the two radial path segments.

$$\oint \mathbf{H}\cdot d\mathbf{L} = \int_A^B H_z\,dz + \cancel{\int_B^C H_\rho\,d\rho} + \int_C^D H_{z,out}\,dz + \cancel{\int_D^A H_\rho\,d\rho} = I_{encl} = \frac{NI}{d}\Delta z$$

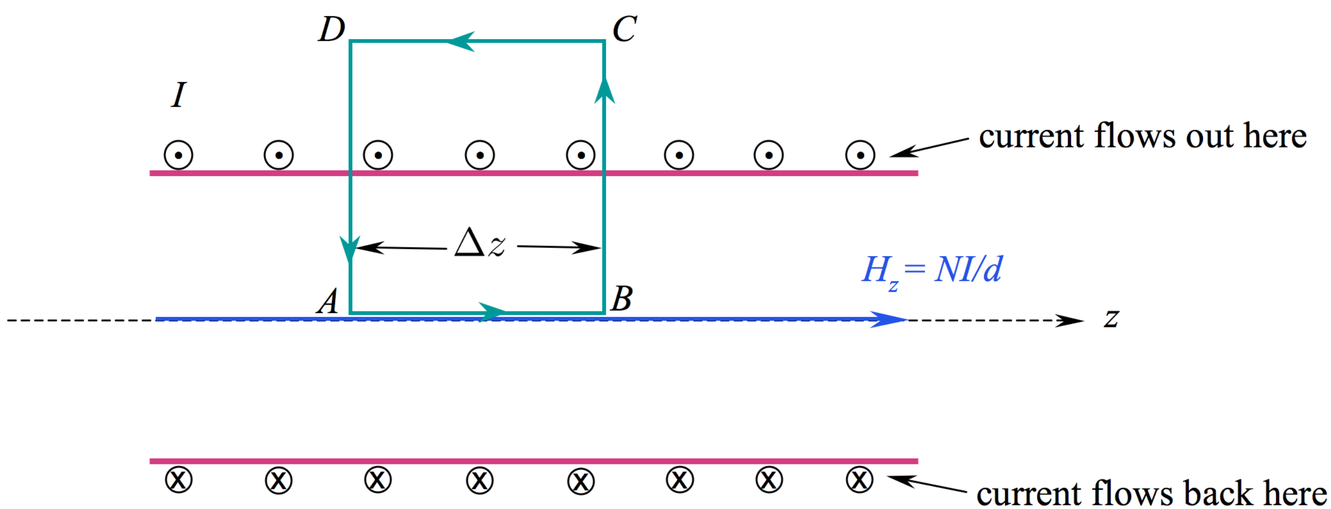


11: The Steady Magnetic Field

- ## Solenoid Field -- Off-Axis

What is left now are the two *z* integrations, the first of which we can evaluate as shown. Since this first integral result is equal to the enclosed current, it must follow that the second integral -- and the outside magnetic field -- are zero.

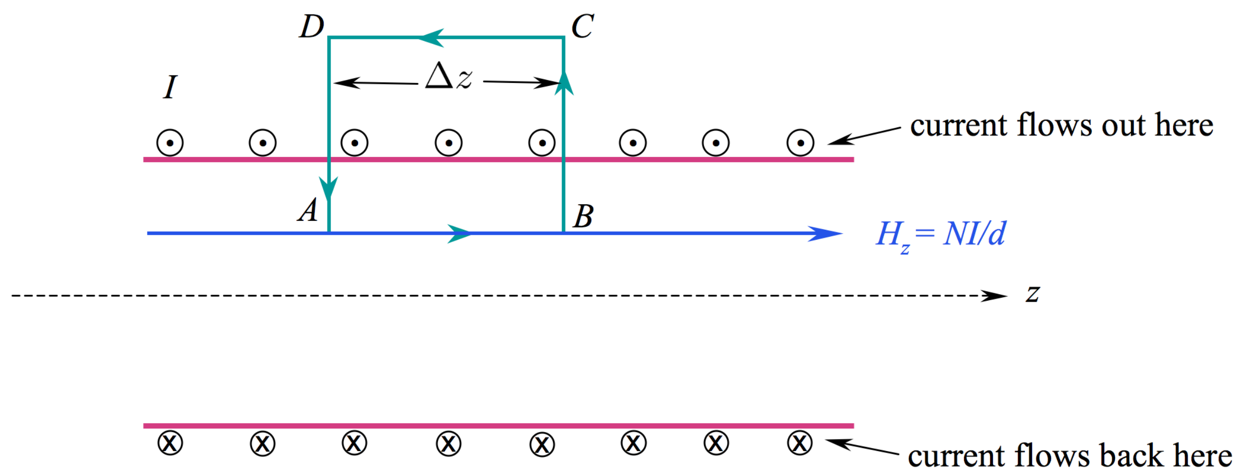
$$\oint \mathbf{H}\cdot d\mathbf{L} = \underbrace{\int_A^B H_z\,dz}_{(NI/d)\Delta z} + \cancel{\int_C^D H_{z,out}\,dz} = I_{encl} = \frac{NI}{d}\Delta z$$



• Solenoid Field -- Off-Axis

The situation does not change if the lower z-directed path is raised above the z axis. The vertical paths still cancel, and the outside field is still zero. The field along the path A to B is therefore NI/d as before.

$$\oint \mathbf{H} \cdot d\mathbf{L} = \underbrace{\int_A^B H_z\, dz}_{(NI/d)\Delta z} + \cancel{\int_C^D H_{z,out}\, dz} = I_{encl} = \frac{NI}{d}\Delta z$$

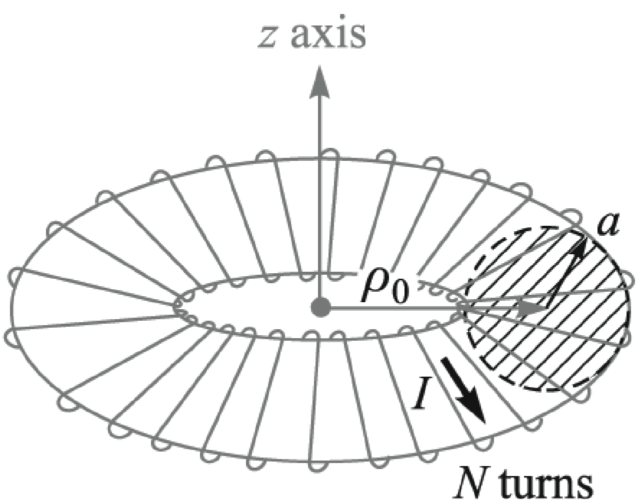


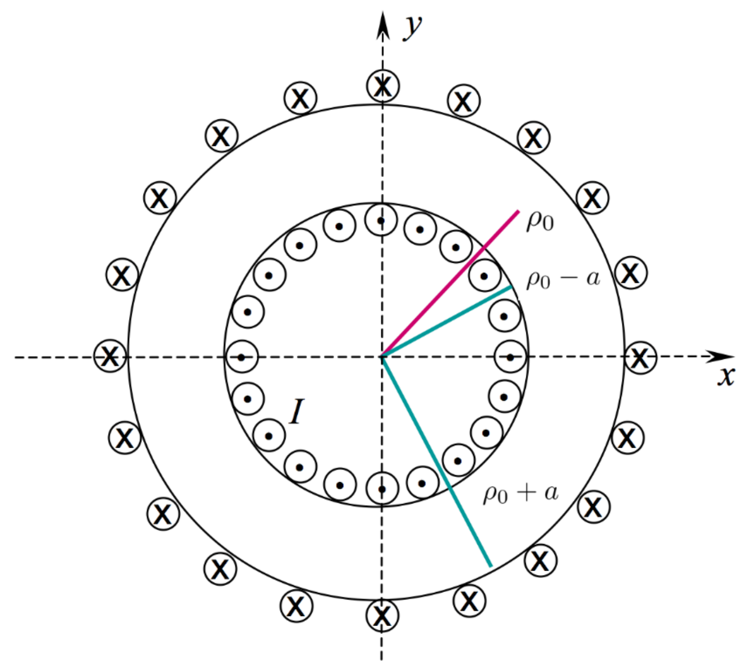
Conclusion: The magnetic field within a long solenoid is approximately constant throughout the coil cross-section, and is $H_z = NI/d$.

- Toroid Magnetic Field

A toroid is a doughnut-shaped set of windings around a core material. The cross-section could be circular (as shown here, with radius a) or any other shape.

Below, a slice of the toroid is shown, with current emerging from the screen around the inner periphery (in the positive *z* direction). The windings are modeled as *N* individual current loops, each of which carries current *I*.





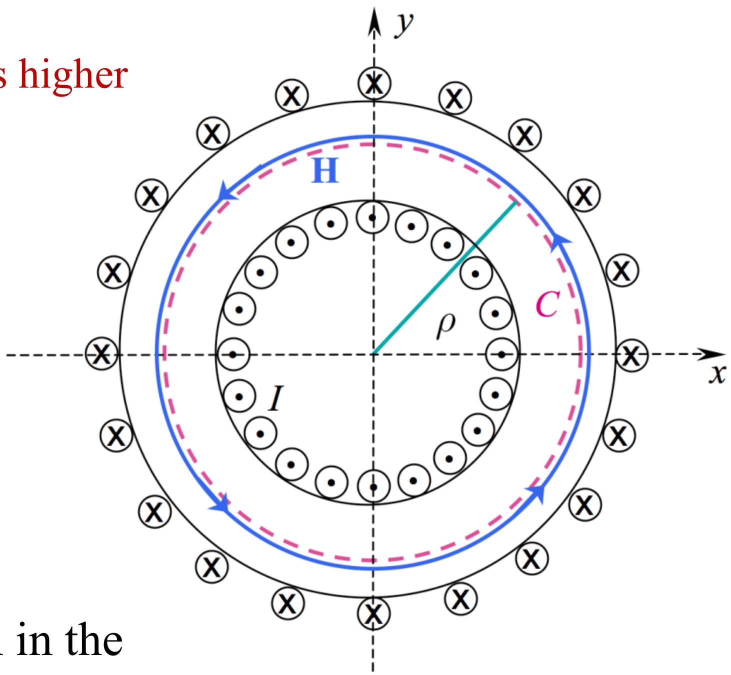
11: The Steady Magnetic Field

- ## Ampere's Law as Applied to a Toroid

Ampere's Circuital Law can be applied to a toroid by taking a closed loop integral around the circular contour C at radius ρ. Magnetic field **H** is presumed to be circular, and a function of radius only at locations within the toroid that are not too close to the individual windings. Under this condition, we would assume:

$$\mathbf{H} = H_\phi \, \mathbf{a}_\phi$$

This approximation improves as the density of turns gets higher (using more turns with finer wire).



Ampere's Law now takes the form:

$$\oint_C \mathbf{H} \cdot d\mathbf{L} = 2\pi\rho \, H_\phi = I_{encl} = NI$$

so that….

$$\boxed{H_\phi = \frac{NI}{2\pi\rho} \qquad (\rho_0 - a < \rho < \rho_0 + a)}$$

Performing the same integrals over contours drawn in the regions $\rho > \rho_0 + a$ or $\rho < \rho_0 - a$ will lead to zero magnetic field there, because no current is enclosed in either case.

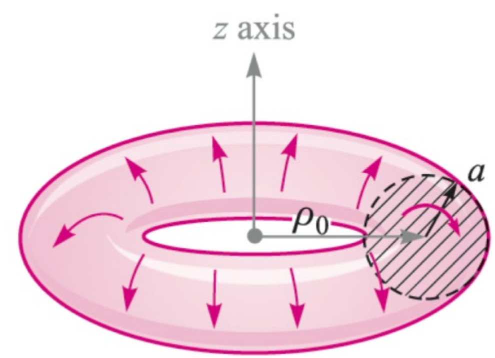
11: The Steady Magnetic Field

- ## Surface Current Model of a Toroid

Consider a sheet current molded into a doughnut shape, as shown. The current density at radius $\rho_0 - a$ crosses the *xy* plane in the *z* direction and is given in magnitude by K_a

Ampere's Law applied to a circular contour *C* inside the toroid (as in the previous example) will take the form:

$$\oint_C \mathbf{H} \cdot d\mathbf{L} = 2\pi\rho H_\phi = I_{encl} = 2\pi(\rho_0 - a)K_a$$



leading to…

$$\boxed{H_\phi = \frac{\rho_0 - a}{\rho} K_a}$$

inside the toroid…. and the field is zero outside as before.